Auditing Closed-Loop Learning in Recurrent Neural Networks: Reproduction, Robustness, and Generalization

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Abstract

Recurrent neural networks are often used as mechanistic models of learning and control, but closed-loop training creates reproducibility challenges because a model's actions alter future inputs. We conduct a claim-level reproducibility study of Ger and Barak's closed-loop RNN learning dynamics, testing independent implementation, seed variation, protocol perturbations, coupled-system diagnostics, and architecture/task transfer. Under a main-text-aligned double-integrator protocol, the trajectory-level peak, not a persistent final gap, reproduces strongly: 50/50 paired seeds show the post-initial open-loop deployed-loss peak, with mean peak/initial ratio 19.1, while final open-loop and closed-loop losses converge after open-loop recovery. The spectral stage and coupled-stability diagnostics also reproduce in 50/50 seeds. A targeted A1 analysis separates stability and behavioral tradeoffs: short-horizon improvements coincide with coupled-radius increases in 120/120 runs; long-horizon loss worsening occurs in 62/120, but never without the radius signal. Generalization is hierarchical: GRU variants preserve final-loss divergence, low-rank variants often produce open-loop deployed rollout blow-ups, tanh RNNs preserve the peak signature without a final gap, tracking transfer is strong, and path-integration transfer is weak. These results motivate reporting practices for closed-loop RNN studies: deployed closed-loop loss, peak signatures, paired seeds, spectral stage criteria, coupled-system spectra, feedback strength, rollout horizon, and failure rates.

1 Introduction

Closed-loop recurrent neural networks occupy an important middle ground between supervised sequence modeling, control, and computational neuroscience. In a closed-loop environment, an RNN's output changes the future inputs it receives. This feedback makes the training problem qualitatively different from open-loop imitation or teacher-forced supervised learning, where future inputs are independent of the learner's deployed actions. Ger and Barak (Ger and Barak, 2025) recently argued that this distinction produces learning dynamics that are fundamentally different from those of otherwise identical open-loop RNNs. Their paper is valuable because it makes a mechanistic claim rather than only a performance claim: the coupled agent-environment system, not the RNN alone, determines important learning transitions.

This paper is not a figure-level reproduction study. Instead, we use Ger and Barak's closed-loop RNN learning framework as a test case for claim-level reproducibility in recurrent control systems. We ask which conclusions survive independent implementation, seed variation, optimizer and hyperparameter perturbations, coupled-system stability analysis, and transfer to nonlinear or path-integration-style settings. This framing matters for TMLR-style reproducibility: a rerun of a public artifact can be educational, but it is only broadly useful when it surfaces reusable lessons about when claims are reliable, when they are protocol-sensitive, and what future papers should report.

Our study makes four contributions. First, we provide an independent, config-driven reproduction of the central open-loop/closed-loop learning comparisons in Ger and Barak. Second, we convert qualitative claims about learning stages and stability transitions into quantitative, seed-level tests with confidence intervals. Third, we measure robustness under perturbations to optimizer, initialization, feedback strength, noise, and rollout horizon, identifying which conclusions are stable and which are protocol-sensitive. Fourth, we distill an audit checklist for future closed-loop RNN studies, including paired-seed comparisons, deployed closed-loop loss, stage criteria, coupled-system stability diagnostics, and failure-rate reporting.

2 Related Work

Task-trained RNNs in computational neuroscience. Task-trained RNNs are widely used as mechanistic models because they can solve cognitive and motor tasks while exposing internal dynamics for analysis. Early and influential work showed that high-dimensional trained RNNs can implement low-dimensional dynamical mechanisms (Sussillo and Barak, 2013), and that recurrent dynamics can account for context-dependent computation in prefrontal cortex (Mante et al., 2013). This line of work has become a general modeling strategy in systems neuroscience, where RNNs serve as hypothesis-generating models rather than only predictive tools (Barak, 2017). At the same time, large-scale analyses of trained recurrent networks show that architecture and training details can change representational geometry and dynamics even when task performance is similar (Maheswaranathan et al., 2019). This motivates our claim-level framing: reproducing a task curve is not enough if the mechanistic conclusion depends on training protocol or model class.

Dynamical-systems analysis of RNNs. The original paper sits in a tradition of reverse-engineering trained RNNs through fixed points, low-dimensional projections, eigenvalues, and linearized dynamics. Sussillo and Barak's low-dimensional analysis made the case that trained recurrent networks can be understood through dynamical-systems tools (Sussillo and Barak, 2013), and later work used related analyses to compare recurrent dynamics across large populations of networks (Maheswaranathan et al., 2019). Our audit differs in emphasis. We do not only ask whether a trained RNN has interpretable internal dynamics; we ask whether the relevant dynamical object is the RNN alone or the coupled agent-environment system. This distinction is central in closed-loop settings because the environment transforms actions into future observations.

Open-loop training, closed-loop deployment, and distribution shift. The mismatch between teacher-forced training and deployed rollouts is well known in sequence prediction and imitation learning. Scheduled sampling was proposed to reduce the discrepancy between training on ground-truth previous tokens and testing on model-generated tokens (Bengio et al., 2015). In imitation learning, DAgger and related reductions explicitly address the fact that a learner's actions change the future states it visits, invalidating i.i.d. supervised-learning assumptions (Ross et al., 2011). Closed-loop RNN learning has an analogous but mechanistically richer mismatch: open-loop imitation can minimize teacher-forced action error while still failing under deployed closed-loop rollouts. This connection motivates our insistence on reporting deployed closed-loop loss, not only open-loop or imitation loss.

Reproducibility and robustness in machine learning. Our audit follows the broader shift from artifact reruns to structured reproducibility studies. The NeurIPS reproducibility program emphasized clear artifacts, documented environments, and reproducibility checklists (Pineau et al., 2021). In deep reinforcement learning, Henderson et al. showed that algorithmic conclusions can be sensitive to seeds, evaluation protocol, and implementation details (Henderson et al., 2018). Closed-loop RNN studies share many of these risks because training outcomes can depend on random initialization, rollout horizon, optimizer, and environment coupling. The present paper therefore treats Ger and Barak's work as a case study for a reusable audit protocol: paired seeds, stress tests, coupled-system diagnostics, and reporting lessons.

We use *reproduction* for tests that preserve the original task family and claim, *robustness* for perturbations of seeds, optimizer, initialization, horizon, noise, feedback, and clipping, and *generalization* for architecture or task-family changes. This terminology is intentionally conservative: a robustness or generalization failure does not by itself falsify the original result, but it does delimit where the claim should be reported as reliable.

Table 1: Direct mapping from original evidence to our reproduction tests. This separates figure-level reproduction from claim-level decisions.

Original claim	Original evidence	Our test	Result	Decision
Closed/open divergence	Loss/gain figures and open-loop sharp peak	50 paired seeds plus peak-signature check	post-initial peak 50/50; final gap 0/50	Reproduced trajectory signature
Stage structure	Spectral stages: negative-position policy, stability phase, refinement	Spectral stage detector plus loss-only changepoints	spectral stages 50/50; segmented loss stages 0/50	Reproduced spectrally
Coupled stability	Coupled eigenvalue argument and Stage 2 alignment	Coupled radius crossing and spectral stage timing	50/50 crossings; median crossing 105.5	Reproduced
Motor-control applicability	Tracking extension	Tracking plus ring variants	tracking 9/10; ring-family 4/50	Partial boundary

3 Original Claims and Audit Questions

We separate the original claims under reproduction from audit questions introduced by this study. The reproduced claims are C1–C3. C1 is the central divergence claim: identical RNNs trained open-loop and closed-loop follow different learning trajectories and differ under closed-loop deployment. C2 is the stage claim: closed-loop learning passes through distinct learning stages. C3 is the mechanistic stability claim: progress depends on stability of the coupled agent-environment system.

The audit questions are A1–A2. A1 asks whether C1–C3 are robust to protocol perturbations and whether a targeted short-vs-long horizon analysis reveals the expected stability tradeoff. A2 asks whether the phenomena generalize across architectures and tasks. We use A1 and A2 to avoid implying that Ger and Barak made every robustness or generalization claim in exactly our framing.

For each reproduced claim or audit question, we define three evidence levels. A result is *reproduced* when the effect and diagnostic hold under the relevant protocol for most paired seeds, with confidence intervals excluding zero or a pre-specified smallest effect size of interest. For final-loss C1 diagnostics this smallest effect is a 5 percent relative deployed-loss gap, corresponding to roughly 0.022 absolute loss in the original-setting closed-loop scale; smaller final gaps are reported continuously but not counted as final-loss divergence. Because the original figure also emphasizes a transient open-loop peak, we report peak-over-training support as a separate C1 signature. A result is *partially reproduced* when the direction appears but is unstable across seeds, sensitive to plausible perturbations, or only weakly supported by the diagnostic. A result is *not reproduced* when the effect is absent, reverses, or depends on hand-picked visual criteria.

4 Closed-Loop RNN Reproducibility Audit Protocol

We specify a modest audit protocol rather than proposing a new benchmark. The protocol is a recipe for testing claims about closed-loop RNN learning dynamics in any recurrent control setting where actions can affect future observations. Its inputs are: a set of original claims, a task/environment, an open-loop training procedure, a closed-loop training procedure, a seed set, a small set of protocol perturbations, and at least one held-out architecture or task variant. Its outputs are: per-seed deployed closed-loop metrics, stage labels, stability diagnostics, perturbation summaries, generalization summaries, and a claim-level reproducibility matrix. Table 2 gives the concrete specification used in this paper.

The protocol deliberately separates *measurement* from *interpretation*. Measurement consists of paired runs, deployed losses, spectra, stage labels, and perturbation outcomes. Interpretation happens only after aggregating seed-level outputs into claim-level decisions. This separation is important because closed-loop RNN papers often make qualitative statements about stages or mechanisms from a small number of curves; our protocol requires those statements to be tied to explicit detectors, uncertainty estimates, and failure rates.

Table 2: Closed-loop RNN reproducibility audit protocol. The protocol is intentionally lightweight: each row defines a required audit step, its minimum implementation, and the output needed to support claim-level conclusions.

Audit step	Minimum requirement	Recorded output
Paired training	Clone the same initialization into open-loop and closed-loop learners; keep architecture and evaluation budget fixed.	Per-seed train loss, deployed closed-loop test loss, peak-over-training deployed loss, final loss, and effective-gain trajectory for both learners.
Seed-level uncertainty	Run enough seeds to estimate direction agreement and confidence intervals; report seed count explicitly.	Mean, 95 percent bootstrap CI, fraction of seeds supporting each claim, and seed-level failure/recovery rates.
Stage detection	Predefine detectors in the same diagnostic space as the claim; for this audit, separate loss-only stages from spectral stages.	Stage labels, plateau duration, transition epochs, spectral-crossing gaps, and frequency of detected stages across seeds/settings.
Coupled diagnostics	Compute RNN-only and coupled agent-environment spectra whenever the system can be linearized or locally linearized.	Spectral radius time series, stability-crossing epoch, plateau-exit gap, and predictive correlation with recovery/failure.
Robustness perturbations	Change factors that alter optimization or environment coupling, such as optimizer, initialization scale, feedback strength, noise, horizon, control penalty, and clipping.	Claim support under each perturbation, effect-size changes, and perturbation-level failure rates.
Generalization check	Test at least one architecture or task variant that preserves action-dependent inputs and one that changes the recurrent/control structure.	Claim support by variant and a boundary-condition statement explaining when the phenomenon does or does not transfer.
Claim/audit decision	Classify each claim or audit question as reproduced, partially reproduced, or not reproduced using the same criteria across settings.	Matrix linking evidence type, original setting, robustness, generalization, and actionable lesson.

4.1 Metrics and run accounting

The main metrics are deployed closed-loop loss, teacher-forced imitation loss, effective feedback gain, coupled and RNN-only spectral radius, algorithmic stage labels, and A1 multi-horizon tradeoff components. Exact formulas are in Appendix B. The main audit contains 455 completed paired runs: 50 core C1–C3 runs, 195 robustness runs, 120 targeted A1 multi-horizon runs, and 90 A2 extension runs; we additionally ran two one-seed protocol sanity checks. Non-finite outcomes are classified as failures (F), not as zero-support evidence. Detector thresholds were selected during smoke-test development before running the full reported sweeps; threshold sensitivity is reported to show which conclusions depend on those choices.

4.2 Paired open-loop and closed-loop training

For each seed, one initialization is cloned into an open-loop and a closed-loop learner. Following Ger and Barak's use of "policy gradients," the closed-loop controller is optimized by taking gradients of the rollout objective with respect to the RNN policy parameters. In this differentiable double-integrator setting, those gradients are computed by backpropagation through the unrolled agent-environment dynamics, matching the direct-gradient setup of the public artifact rather than REINFORCE-style stochastic likelihood-ratio estimation. The open-loop model imitates teacher trajectories in our independent implementation. Both are evaluated under deployed closed-loop rollout.

4.3 Stage detection

The original paper anchors its stages spectrally: Stage 1 is a negative-position policy with an unstable complex pair, Stage 2 ends when the dominant eigenvalues of the coupled matrix enter the unit disk, and Stage 3 involves policy refinement and growth of a third real mode. Our main C2 detector therefore operates in the same spectral space: it records the effective position gain, the unstable complex coupled mode, the coupled stability crossing, and post-crossing growth of the third real mode. We also report loss-only derivative and segmented-changepoint observers as downstream diagnostics, not as substitutes for the spectral claim.

4.4 Coupled-system stability diagnostics

For linearized settings, we compute the spectral radius of the RNN recurrent matrix and of the coupled agent-environment transition matrix. This targets C3 directly: if closed-loop dynamics are governed by the agent-environment loop, the coupled diagnostic should be more informative than an isolated RNN diagnostic.

4.5 Robustness and generalization

The A1 robustness sweep changes learning rate, initialization scale, feedback strength, episode length, observation noise, control penalty, optimizer, and gradient clipping. The Adam condition is included because Ger and Barak's Appendix C.7 predicts that adaptive optimization mitigates the structured short/long-horizon conflict. The A2 generalization sweep tests nonlinear RNNs, GRUs, low-rank RNNs, tracking control, and ring/path-integration-style tasks.

4.6 Original artifact, independent implementation, and deviations

We preserved the public artifact under `external/original_artifact/`. It contains the original notebooks for figure reproduction, nonlinear training, theory, and tracking, plus the pickle files used by the artifact. Our reproduction log records which files load, which notebooks depend on precomputed data, and which notebook runs are long or checkpoint-heavy.

The experiments reported here use an independent, config-driven implementation rather than direct notebook execution. The double-integrator task, paired open/closed training, effective gains, coupled spectra, stage detectors, robustness perturbations, and generalization tasks were reimplemented as package modules and command-line scripts. We matched the original double-integrator architecture and training profile where needed for C1–C3, then intentionally introduced new A1/A2 audits over seeds, optimizer, initialization, feedback, horizon, noise, clipping, architecture, and task. We did not contact the original authors; all choices were made from the paper, public artifact, and documented configs. Exact notebook reruns are therefore treated as artifact checks, while the paper's quantitative claims are evaluated through the independent implementation. Every table and figure can be regenerated from saved raw CSVs with `scripts/run_full_audit.sh`, `scripts/run_targeted_c2_a1_a2.sh`, and `python -m closed_loop_repro.plotting.make_all_figures -config configs/figures/tmlr.yaml`.

A code-level comparison shows that the core reproduction matches the original double-integrator dynamics, hidden size, tanh controller class, SGD learning rate, batch size, rollout horizon, gradient clipping, and teacher-student design. It is not a line-for-line reimplementation: the original nonlinear notebook uses a zero first action, current-state loss accumulation, white-noise teacher inputs for open-loop training, 1001 epochs, smaller input/output initializations under its small-scale setting, and a small control penalty. The main-text-aligned config used for the reported core run sets $\beta = 0$, samples $x_0 \sim [-2, 2]^2$, averages the squared Euclidean state cost over the rollout, and keeps the notebook's timestep ordering. We therefore interpret the package as an independent claim-level audit of the original protocol family, not an exact numerical rerun of every notebook.

This protocol alignment changed the interpretation of C1. During development, an open-loop implementation that did not use the paper/artifact white-noise teacher-input convention produced a persistent final deployed-loss gap. After correcting that mismatch, the faithful protocol shows the original transient open-loop peak

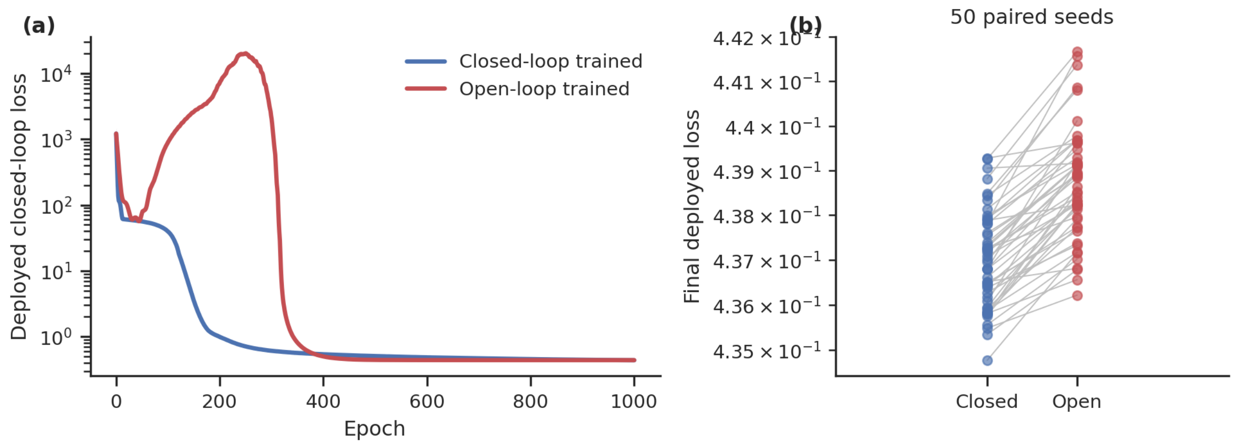

Figure 1: Core C1 reproduction under the original double-integrator setting. (a) Median deployed closed-loop test loss across paired seeds, with interquartile bands. The open-loop student shows the post-initial deployed-loss peak in 50/50 seeds. (b) Paired final deployed losses for the same initializations. Final losses are similar after open-loop recovery, so the reproduced C1 evidence is the trajectory-level peak rather than a large final gap.

followed by final convergence. We therefore report C1 with two sub-diagnostics throughout: the peak-over-training signature and the final deployed-loss gap.

5 Result 1: Direct Reproduction of the Core Divergence

The original-setting double-integrator runs reproduce the trajectory-level C1 signature (Figure 1). Across 50 paired seeds, every open-loop student exhibits a post-initial deployed-loss peak. The peak occurs at mean epoch 249.1 with bootstrap 95 percent CI $[247.5, 250.5]$ and median epoch 249.5. The mean peak/initial ratio is 19.1 with CI $[17.1, 21.4]$, and the mean peak/final ratio is 4.88×10^4. This is the qualitative open-loop failure mode emphasized in the original figure: teacher-forced imitation can look successful while deployed closed-loop loss transiently explodes.

Final losses converge after the open-loop recovery. The closed-loop trained model reaches final deployed loss 0.4370 with bootstrap 95 percent CI $[0.4367, 0.4373]$, seed SD 0.00109, and IQR $[0.4361, 0.4378]$. The open-loop trained student has final loss 0.4387 with CI $[0.4383, 0.4390]$, seed SD 0.00126, and IQR $[0.4379, 0.4393]$. The paired final gap is positive but small, 0.00168 with CI $[0.00147, 0.00190]$, and does not exceed our 5 percent final-loss threshold in any seed. Thus C1 reproduces as a learning-trajectory divergence and peak-over-training signature, not as a persistent final deployed-loss separation.

The spectral C2 and C3 signatures also appear in the core run. The spectral three-stage detector is supported in 50/50 seeds, with median Stage 1 end at epoch 7 and median Stage 2/stability crossing at epoch 105.5. Coupled stability crossings occur in 50/50 seeds, and the final coupled spectral radius is 0.6253 with CI $[0.6243, 0.6264]$.

6 Result 2: Robustness and Tradeoff Audit

The robustness sweep consists of 195 paired runs: 13 perturbation settings with 15 seeds per setting (Table 3 and Figure 3). The peak-signature version of C1 is robust across finite SGD-like perturbations: it appears in 165/195 robustness runs, i.e. all finite settings except Adam. The final-loss version is more selective, supported in 60/195 runs. Those 60 successes are exactly the low-learning-rate, high-learning-rate, short-horizon, and Adam rows (15/15 each). The baseline robustness row is 0/15 for final-loss C1, matching the 0/50 core run; long horizon, feedback, initialization, noise, and high-control-penalty rows are also 0/15

Table 3: Robustness perturbation grid. The original protocol also has the separate 50-seed core reproduction; the robustness grid repeats the baseline for 15 additional paired seeds so every perturbation has the same seed count.

Perturbation	Changed value	Seeds
`original`	Baseline double-integrator protocol	50 core; 15 grid
`lr_low`	learning rate 0.003	15
`lr_high`	learning rate 0.03	15
`init_weak`	recurrent initialization scale 0.03	15
`init_strong`	recurrent initialization scale 0.3	15
`feedback_low`	environment feedback strength 0.5	15
`feedback_high`	environment feedback strength 1.5	15
`episode_short`	rollout horizon 25 steps	15
`episode_long`	rollout horizon 100 steps	15
`obs_noise`	observation noise standard deviation 0.05	15
`adam`	optimizer Adam with learning rate 0.001	15
`high_control_penalty`	control penalty 0.05	15
`no_clip`	gradient clipping disabled	15

because open-loop recovery erases the final gap. Adam is therefore not a simple weakening of all closed/open signatures: it removes the transient peak (0/15) but leaves a persistent final deployed gap (15/15), suggesting an optimizer-specific failure mode rather than the SGD-like peak-and-recovery trajectory. With gradient clipping disabled, all 15 `no_clip` runs produce non-finite final losses and are counted as failures rather than negative evidence for the claim.

The stage result splits by diagnostic (Figure 2). The spectral C2 detector is supported in 180/195 robustness runs, failing only when training becomes non-finite without clipping. Loss-only stage detectors are more diagnostic-dependent: the derivative-style loss detector finds three-stage structure in many finite runs, but segmented changepoints on the original loss curves do not recover the same boundaries. This supports the reporting lesson that stage claims should be tied to the diagnostic space in which they are defined.

The coupled-system stability transition is robust wherever the coupled spectrum is available and the run remains finite. It is supported in 180/195 robustness runs, again failing only in the no-clipping condition. This supports the original mechanistic emphasis in linearizable settings and separates spectral stability from downstream loss-changepoint sensitivity.

The targeted A1 sweep is more specific than the original union metric suggests (Figure 4). Across six control penalties and 20 seeds per penalty, the union criterion–short-horizon improvement plus either long-horizon loss worsening or increased coupled spectral radius–is supported in 120/120 runs, with mean conditional fraction 0.4724. Splitting the criterion shows that the radius signal is universal: short-horizon improvement plus increased coupled radius is supported in 120/120 runs. Long-horizon loss worsening occurs in 62/120 runs, and the intersection of loss worsening and radius increase is also 62/120. The exclusive loss-only criterion is 0/120, while the exclusive radius-only criterion is 120/120. This distinction prevents the audit from overstating a purely behavioral loss tradeoff: every detected loss-worsening case is accompanied by the stability-cost signal.

7 Result 3: Which Diagnostics are Predictive?

The central C2 result is diagnostic-specific. The spectral detector reproduces the original stage structure in 50/50 core seeds: negative position gain appears early, an unstable complex coupled mode is present, the dominant coupled eigenvalues enter the unit disk near epoch 105.5, and the third real mode subsequently grows. In contrast, segmented regression on deployed loss alone finds median changepoint boundaries at epochs 120 and 193 with slopes -0.0117, -0.0432, and -0.00065. This loss-only observer does not show

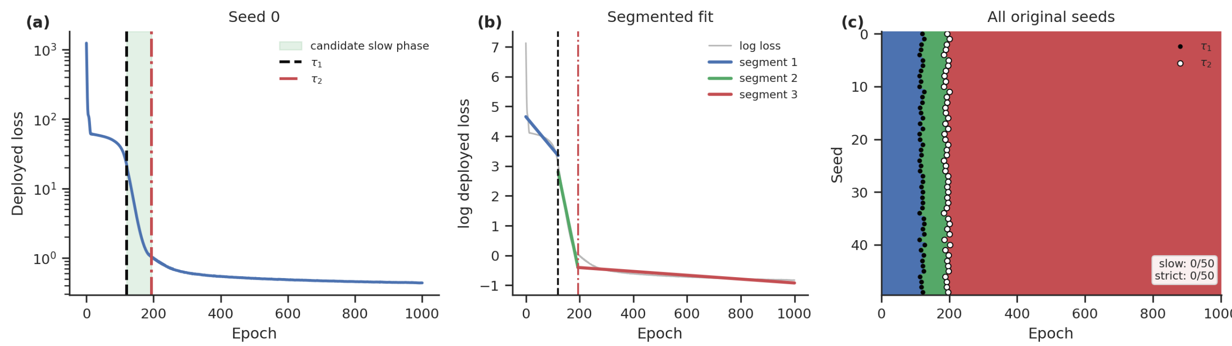

Figure 2: Stage analysis for the original-setting runs using downstream loss changepoints. (a,b) A representative seed with candidate boundaries τ_1 and τ_2 and segmented fits. (c) Aggregate stage raster for all 50 original seeds; black dots mark τ_1 and white dots mark τ_2. The segmented detector uses minimum segment length 20 epochs, Stage 1 slope < -0.02, Stage 2 slope magnitude below $\max(0.35|\text{slope}_1|, 0.004)$, and Stage 3 reacceleration slope $< \text{slope}_2 - 0.002$. Unlike the spectral detector, this loss-only segmented observer does not recover robust strict three-stage support.

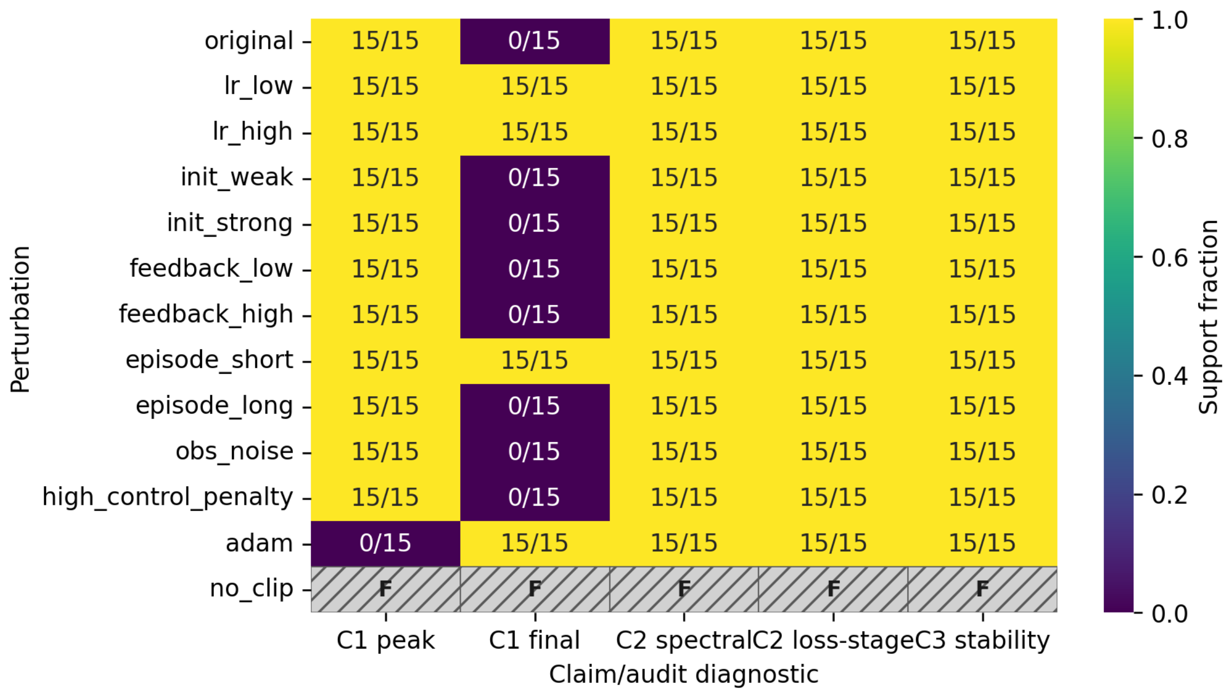

Figure 3: Robustness heatmap over perturbation settings and claim diagnostics. C1 is split into the transient peak signature and the final deployed-loss criterion, so the original row correctly shows peak support but no final-gap support. C2 is split into spectral support and a broader downstream loss-stage diagnostic. A1 is omitted here because the coarse robustness proxy is not the targeted multi-horizon A1 test in Figure 4. Disabling gradient clipping produces non-finite failures marked as F rather than zero support.

the same stage ordering under any threshold variant in Table 4. Thus C2 is reproduced spectrally, while loss-only visual stage boundaries are not a reliable substitute for spectral criteria.

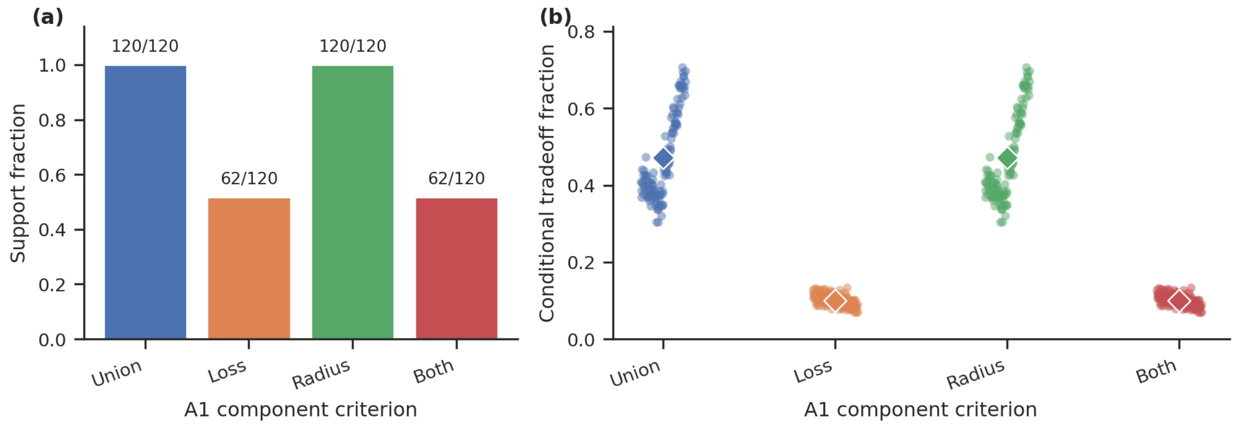

Figure 4: Targeted A1 component analysis over 120 paired tradeoff runs. (a) Support counts for the union criterion, long-horizon loss-worsening criterion, radius criterion, and the intersection of loss and radius criteria. (b) Seed-level conditional fractions among short-horizon improvement intervals. The union and radius criteria are supported in 120/120 runs; loss worsening appears in 62/120 and never without the radius signal. Diamonds show means.

Table 4: C2 threshold sensitivity for segmented loss-only changepoints on the 50 original-setting seeds. The spectral C2 detector is supported in 50/50 seeds, but this downstream loss-only observer does not recover the original stage ordering across looser, stricter, or shorter-segment variants.

Detector variant	**Slow phase**	**Strict three-stage**	**Median** τ_1, τ_2
Main segmented	0/50	0/50	120, 193
Loose segmented	0/50	0/50	120, 193
Strict segmented	0/50	0/50	120, 193
Short-segment binary style	0/50	0/50	120, 193
Very loose reacceleration	0/50	0/50	120, 193

The original-setting coupled-system stability transition is detected in 50/50 seeds (Figure 5). The final coupled spectral radius is finite and below the stability threshold in every original run, and the median crossing epoch is 105.5. Because the spectral C2 detector defines the Stage 2 boundary by this crossing, the substantive timing test is whether the crossing aligns with the learning plateau exit. It does under the derivative-style plateau detector: the median stability-to-plateau gap is one epoch. Segmented loss-only changepoints give different downstream boundaries, reinforcing that C3 should be evaluated in coupled spectral space rather than from visual loss-curve annotations alone.

The spectral diagnostic is architecture-dependent in the current tooling, not in principle. For linear, tanh, and low-rank controllers on linear tasks, the coupled transition matrix can be constructed explicitly. Tracking uses a different coupled-matrix construction from the double integrator, which makes its 3/10 C3 support a weaker diagnostic transfer result. We do not report coupled spectral diagnostics for GRU or ring variants because this implementation does not yet compute local Jacobian spectra for gated nonlinear controllers. This is a limitation of the audit code: local linearization should be added before making claims about C3 transfer to GRUs or nonlinear task families.

8 Result 4: Does the Phenomenon Generalize?

Generalization is hierarchical rather than uniform (Figure 6). Across 90 completed architecture/task extension runs, final deployed-loss divergence is supported in 33/90 runs, while the open-loop peak signature appears in 56/90. Architecture transfer within the double-integrator-style control family depends on the diagnostic: GRU variants show final-loss divergence in 10/10 seeds, and low-rank variants also pass the final-loss criterion in 10/10 seeds but do so through open-loop deployed rollout blow-ups rather than cal-

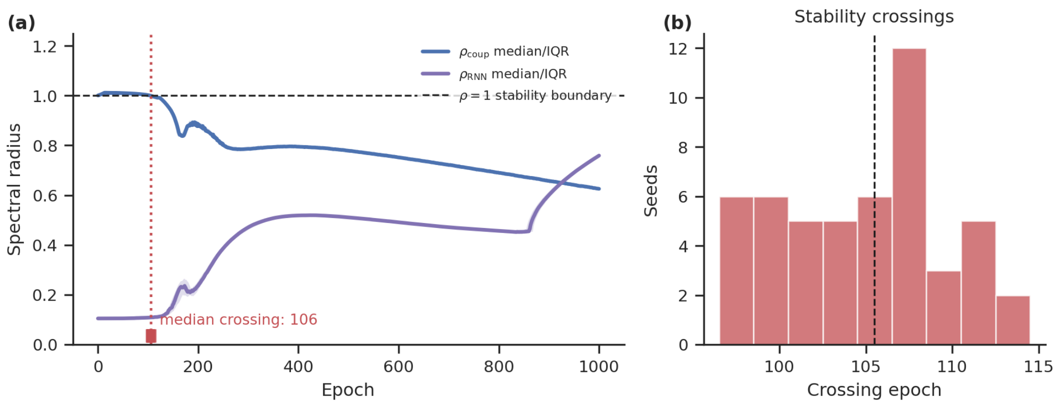

Figure 5: Coupled-system spectral analysis for the original setting. (a) Median coupled and RNN-only spectral radii across 50 seeds, with interquartile bands; the dashed line marks the $\rho = 1$ stability boundary and tick marks show per-seed crossing epochs. The plotted median marker is rounded to epoch 106; the unrounded median used in the text is 105.5. (b) Distribution of stability-crossing epochs. The coupled spectral radius crosses below the stability boundary early in training, whereas the RNN-only spectral radius remains far from the relevant stability threshold.

ibrated final losses. In the low-rank condition, finite open-loop final losses range from 7.8 to 1.9×10^{13}; Figure 6 clips displayed gaps above 10^3 and marks them with triangles. Tanh RNNs preserve the peak signature in 10/10 but converge to nearly identical final losses. Task transfer to tracking is strong under the final-loss criterion, with deployed-loss divergence in 9/10 seeds. Task-family transfer to the harder partial-observation moving-target ring variants is weak, with 3/20 GRU seeds and 1/20 tanh seeds passing the deployed-loss criterion.

The simple ring variant is best viewed as a sanity check, not as a hard generalization test: it gives the learner sin(error), cos(error), and velocity with a fixed target, making the mapping nearly memoryless. As expected, it has 0/10 supporting seeds and mean deployed-loss gap -1.3×10^{-4}. The harder ring follow-up is fairer but still weak, with mean deployed-loss gaps 0.0837 for GRU and 0.0085 for tanh. These results sharpen A2: the closed/open divergence transfers across some architecture and tracking changes, but it does not automatically transfer to path-integration variants.

C3 diagnostic transfer is only tested where the coupled spectrum is implemented. It is supported in 10/10 tanh RNN runs and 10/10 low-rank runs, and in 3/10 tracking runs, with the caveat that tracking uses a different coupled-matrix construction than the double-integrator case. It is not reported for GRU or ring runs because the current spectral code does not linearize gated nonlinear controllers. Local Jacobian spectra for gated nonlinear controllers are needed before claiming diagnostic transfer to GRUs. This is another reporting lesson: a generalization claim about stability mechanisms should distinguish performance transfer from diagnostic transfer.

9 Claim-Level Audit Matrix

Table 5 is the main summary of the paper. It separates original claims C1–C3 from audit questions A1–A2, and attaches each to evidence, robustness or generalization results, and an actionable lesson.

10 Actionable Lessons for Closed-Loop RNN Studies

The goal of the audit is to produce reusable reporting practices. We propose the following checklist for closed-loop RNN papers.

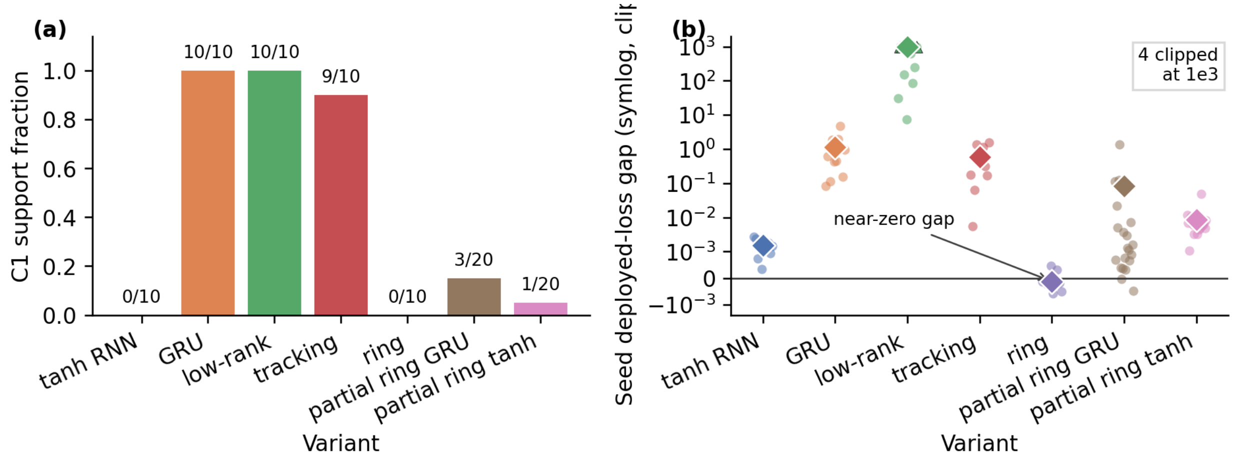

Figure 6: Generalization results across architecture and task variants. Standard variants use $n = 10$ paired seeds; the harder partial-observation ring variants use $n = 20$ paired seeds. (a) C1 final-loss support fractions with exact seed counts; tanh RNN is 0/10 here but preserves the open-loop peak criterion in 10/10 seeds. (b) Seed-level deployed-loss gaps with mean diamonds on a symmetric-log scale, clipped at 10^3 for display; triangle markers indicate clipped low-rank open-loop rollout blow-ups. The simple ring is a sanity check for a nearly memoryless setting; the partial-observation ring variants are the stronger path-integration tests.

Table 5: Claim-level audit matrix for completed runs. C1–C3 are original claims under reproduction. A1–A2 are audit questions introduced here to test robustness, short/long horizon tradeoffs, and broader generalization.

Item	Evidence type	Original protocol	A1 audit	A2 audit	Lesson
C1: open/closed divergence	Final deployed loss and peak signature	Reproduced peak 50/50; final gap 0/50	Peak robust in finite SGD-like settings; final gap diagnostic varies; no clipping fails	GRU/low-rank/tracking final gaps, tanh peak only, ring weak	Report deployed loss and peak-over-training signatures.
C2: stage structure	Spectral detector and loss-only observers	Spectral C2 50/50; segmented loss stages 0/50	Spectral C2 robust in finite linearizable settings	Spectral transfer only where matrix is implemented	Stage claims need diagnostics in the same space as the claim.
C3: coupled stability	Coupled spectral radius and timing	Crossing 50/50; median epoch 105.5	Robust finite crossings	Holds for tanh/low-rank; local Jacobians not implemented for GRU/ring	Report coupled spectrum and its timing relative to stage boundaries.
A1: protocol robustness and tradeoff	Perturbation grid and multi-horizon component metric	N/A	Union/radius 120/120; loss worsening 62/120; exclusive loss-only 0/120	N/A	Robustness claims need stress tests, explicit horizons, and component criteria.
A2: broader generalization	Architecture and task extensions	N/A	N/A	Partial: diagnostic-dependent architecture transfer, tracking strong, ring weak	Closed-loop effects depend on task structure, environment coupling, and observability.

Report deployed closed-loop performance and peak-over-training signatures. The central risk is that a model can imitate a teacher under fixed inputs but fail when its own actions determine future inputs; final loss alone can miss transient peaks that define the qualitative phenomenon.

Use paired-seed comparisons when contrasting open-loop and closed-loop training. The clean comparison uses the same initialization, architecture, optimizer budget, and evaluation rollouts while changing only the training regime.

Define learning stages in the diagnostic space used by the claim. Spectral stage claims should report eigenvalue criteria, order parameters, and boundary epochs, while loss-only changepoints should be labeled as downstream observables.

Analyze the coupled agent-environment system. RNN-only spectra can miss stability changes introduced by the environment. Coupled diagnostics should be reported whenever a mechanistic stability claim is made.

Report feedback strength and rollout horizon. These parameters determine how strongly actions affect future observations and can control whether the closed-loop phenomenon appears.

Report failure rates, not just average curves. Means can hide unstable runs, recovery failures, or seed-dependent plateaus.

Include at least one stress test. Optimizer, initialization, noise, feedback strength, and horizon perturbations are inexpensive compared with the interpretive cost of an untested mechanistic claim.

11 Discussion

This project treats Ger and Barak's framework as a case study in reproducibility for interactive recurrent systems. The completed audit supports the central trajectory-level open/closed divergence, spectral stage structure, and coupled stability transition under a main-text-aligned double-integrator protocol. It also identifies important limits. First, final deployed loss is not the right sole diagnostic for C1: the original-setting final gap is small after open-loop recovery, while the transient open-loop peak is robust. Second, spectral C2 reproduces, but segmented loss-only changepoints do not recover the same stage ordering. Third, A1 shows a stability-cost signal through coupled-radius increases, and long-horizon loss worsening appears in about half of runs but never without that radius signal. Fourth, generalization is hierarchical and diagnostic-dependent: some architecture and tracking variants transfer, while path-integration variants mostly do not.

These mixed results are the reason a claim-level audit is useful. A figure-level reproduction might emphasize only that a curve looks similar for one seed. A methodological audit can say more: final and peak diagnostics tell different stories, missing gradient clipping can produce outright failure, optimizer and horizon choices change whether the final gap persists, path-integration-style variants can erase the gap, spectra for gated controllers require local Jacobian tooling, and stage claims should be accompanied by spectral boundaries rather than visual annotations alone. This is especially important for computational neuroscience, where RNNs are often used as mechanistic models and where closed-loop behavior is closer to biological learning than fixed-dataset supervision.

12 Limitations

The audit does not cover every RNN architecture, every control task, or full reinforcement-learning algorithms. The policy-gradient setting is deterministic and differentiable: training uses direct gradients through the rollout, not sparse-reward or likelihood-ratio RL. The path-integration extensions are intentionally lightweight, and the simple ring variant is a sanity check for a nearly memoryless setting. The A1 tradeoff metric is interval-level and diagnostic: it shows that short-horizon improvement often coincides with increased coupled spectral radius; long-horizon loss worsening is present in some settings but not as an exclusive loss-only signal. Finally, coupled spectral analysis is implemented for explicitly constructed coupled matrices, but not yet for local Jacobians of GRUs or other gated nonlinear controllers.

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A Artifact Entry Points

The accompanying anonymous repository provides a config-driven implementation with smoke and full-run modes, raw CSV outputs, claim tables, and figure-generation scripts. The artifact is organized around the audit protocol rather than notebook-only reproduction: each experiment records the config, seed, device, metrics, and per-epoch time series. During double-blind review, the supplementary archive omits public repository URLs, Software Heritage origin links, Git remotes, commit history, local logs, and machine-specific paths; a public archival link should be added only after de-anonymization. Table 6 lists the main entry points.

Table 6: Artifact entry points. Full runtimes are approximate for one NVIDIA L40 GPU and are intended for planning rather than benchmarking.

Component	**File/script**	**Produces**	**Runtime**	**Expected output**
Smoke validation	`scripts/run_smoke.sh`	Tiny raw outputs, tables, figures	$<$ 10 min	`results/raw/smoke_*`
Full audit	`scripts/run_full_audit.sh`	Core, robustness, generalization, figures	16–18 h	sweep summaries, figures
Targeted C2/A1/A2	`scripts/run_targeted_c2_a1_a2.sh`	Tradeoff, hard ring, changepoints	8–10 h	targeted summaries
Claim tables	`make_claim_tables` module	Claim matrix	$<$ 1 min	`claim_matrix.csv`
Supplemental tables	`supplemental_audit_tables` module	Run accounting, C2 sensitivity, A1 split, peak/alignment checks	$<$ 1 min	processed CSVs
Figures	`make_all_figures` module	Paper figures	$<$ 1 min	`figure_*.png`

B Metric Definitions

Let s_t be the environment state, $o_t = Cs_t + \epsilon_t$ the observation, h_t the recurrent state, and $u_t = f_\theta(o_t, h_t)$ the action. Closed-loop deployment follows $s_{t+1} = F(s_t, u_t)$ and $h_{t+1} = G_\theta(o_t, h_t)$. The deployed loss is

$$L_{\text{deploy}}(\theta) = \frac{1}{T}\sum_{t=0}^{T-1}\left[\ell_s(s_t, s_t^\star) + \lambda_u\|u_t\|_2^2\right], \tag{1}$$

while teacher-forced imitation loss compares the learner action to the teacher action on teacher-generated observations:

$$L_{\text{TF}}(\theta;\theta^\star) = \frac{1}{T}\sum_{t=0}^{T-1}\|f_\theta(o_t^\star, h_t) - f_{\theta^\star}(o_t^\star, h_t^\star)\|_2^2. \tag{2}$$

For linearizable double-integrator settings we estimate an effective feedback gain by ridge regression, $\widehat{K}_t = (X^\top X + \alpha I)^{-1}X^\top U$, and report the closed/open gain distance $D_K = (\sum_{t\in\mathcal{E}}\|\widehat{K}_t^{\text{CL}} - \widehat{K}_t^{\text{OL}}\|_F^2)^{1/2}$.

For linear RNN controllers and linear environments, with $s_{t+1} = As_t + Bu_t$, $u_t = W_{ho}h_t$, and $h_{t+1} = (1-\eta)h_t + \eta(W_{hh}h_t + W_{ih}Cs_{t+1})$, the coupled transition matrix over $z_t = [s_t^\top, h_t^\top]^\top$ is

$$P_\theta = \begin{bmatrix} A & BW_{ho} \\ \eta W_{ih}CA & (1-\eta)I + \eta W_{hh} + \eta W_{ih}CBW_{ho}\end{bmatrix}. \tag{3}$$

We report $\rho_{\text{RNN}} = \max_i|\lambda_i(W_{hh})|$ and $\rho_{\text{coup}} = \max_i|\lambda_i(P_\theta)|$. The stability-crossing epoch is the first epoch at which $\rho_{\text{coup}} < 1$ for a minimum persistence window.

The open-loop peak signature is evaluated over the deployed-loss trajectory across training epochs. We count a post-initial peak when $\arg\max_e L_{\text{deploy},e}^{\text{OL}} > 0$ and $\max_e L_{\text{deploy},e}^{\text{OL}} > 1.5L_{\text{deploy},0}^{\text{OL}}$; we also report peak/initial and peak/final ratios.

The spectral stage detector follows the original diagnostic structure. It records Stage 1 support when the effective position gain is negative and the coupled system has a persistent unstable complex mode; Stage 2 support when the dominant coupled radius persistently enters the unit disk; and Stage 3 support when the largest real coupled eigenvalue grows after that stability crossing. For comparison, loss-only labels use smoothed log deployed loss $y_t = \log(\max(L_{\text{deploy},t}, \varepsilon))$. The segmented loss variant fits three piecewise-linear segments and requires Stage 1 slope < -0.02, Stage 2 slope magnitude below $\max(0.35|\text{slope}_1|, 0.004)$, and Stage 3 slope $< \text{slope}_2 - 0.002$.

For A1, define a myopic-improvement interval by $\Delta\log L_e^{(T_s)} < -\delta_{\text{short}}$. We report four component criteria: union $I_s \wedge (I_\ell \vee I_\rho)$, loss $I_s \wedge I_\ell$, radius $I_s \wedge I_\rho$, and both $I_s \wedge I_\ell \wedge I_\rho$, where I_ℓ is $\Delta\log L_e^{(T_\ell)} > \delta_{\text{long}}$ and I_ρ is $\Delta\rho_{\text{coup},e} > 0$. In the targeted sweep, $T_s = 10$, $T_\ell = 200$, and $\delta_{\text{short}} = \delta_{\text{long}} = 0.005$.

For scalar paired metric M, the open/closed effect is $\Delta_i(M) = M_i^{\text{OL}} - M_i^{\text{CL}}$. We report mean, nonparametric bootstrap 95 percent CI, seed-level SD/IQR where useful, direction agreement $N^{-1}\sum_i \mathbf{1}[\Delta_i(M) > 0]$, and binary claim-support rate $S(c) = N^{-1}\sum_i q_i(c)$.